MILESTONES

1. **PROJECT TRIGGER & PROBLEM DEFINITION**
   Identifying the tension between precision and ritual in herbal tea preparation during daily routines.
2. **DESK & FIELD RESEARCH**
   Identifying the tension between precision and ritual in herbal tea preparation during daily routines.
3. **USER RESEARCH & DATA STRUCTURİNG**
   Organizing interview transcripts, observation notes, and group discussions using AI to identify recurring behavioral patterns.
4. **AI PSYCHOLOGICAL REFLECTION LOG (DÜŞ)**
   Using AI as a reflection tool to document psychological blocks and reorganize the design process during periods of confusion.
5. **USER JOURNEY AND SCENARIO**
   Mapping the tea preparation process to identify friction points in dosage estimation, brewing time, and cleaning.
6. **PRODUCT CONCEPT**
   Defining a guided semi-automated tea brewing system that ensures accurate steeping while preserving the preparation ritual.
7. **INTERFACE**
   Designing a guided interface that displays brewing steps, ingredient guidance, and steeping control.
8. **SKETCHING & FORM EXPLORATION**
   Generating early design sketches to explore product form, brewing chamber visibility, and interaction points.
9. **PRODUCT IDEA**
   Presenting the final design through 3D models, render visuals, and interaction scenarios.

PROJECT TRIGGER + PROBLEM DEFINITION

Milestone 01 – Problem Discovery

Preparing herbal tea blends requires correct proportions and steeping time. However, during daily routines users often rely on approximation and memory. When attention shifts to other activities, tea can easily become over-steeped, leading to bitterness, incorrect dosage, or inconsistent taste.

In herbal tea preparation, dosage also matters because some plants require controlled quantities. At the same time, fully automated machines remove users from the preparation ritual, which reduces engagement and trust in the process.

This project explores how a brewing system can balance precision and ritual during everyday tea preparation.

REFLECTION	KEYWORDS	SELECTED PROMPT
Project scope exploration. This prompt helped map the beverage landscape and define the project scope.	• Precision • Ritual • Steeping control • Daily routines	“List me all the hot drinks. I want you to include local drinks from Turkish culture in the list.”

AI ROLE	AI RESPONSE SUMMARY	DESIGN DECISION
Clarifier	AI generated a broad list of hot beverages including tea, herbal infusions, coffee, hot chocolate, and culturally specific drinks.	Focus on tea and herbal tea preparation instead of designing a general hot beverage machine.

Accepted: Tea and herbal tea preparation systems as the core focus.
Rejected: Soup preparation machines as a hot beverage product direction.
Transformed: Hot beverage machine → tea preparation system.
After searching online and dicussing with a friend if soup is a drink or not.

RESEARCH

Milestone 02- Interview, Desk, Field

Desk research examined tea consumption patterns, household structures, and tea preparation culture.
In Türkiye, tea preparation is strongly connected to daily routines and shared domestic environments. Multi-person households are common, which means tea preparation often becomes a shared activity.
Tea is also a growing global market. Herbal tea consumption is increasing due to health perception and interest in natural beverages.
Field research included observation sessions in a kitchen environment where people prepared hot beverages. During these sessions, user actions, mistakes, and preparation habits were recorded.
Users often rely on visual cues such as tea color instead of precise steeping time.

REFLECTION

The research material included transcripts, observation notes, and informal discussions. AI helped structure the information.

KEYWORDS

- Tea ritual
- Daily routine
- Health perception
- Functional beverage

SELECTED PROMPT

"Help me organize the insights from my interviews and observation notes + transcripts. I coded attendees for privacy."

Insight 1 Users rely on tea color rather than timing.
Insight 2 The hidden problem is steeping time and control.
Insight 3: Cleaning and limescale are critical usability concerns.

AI ROLE

- Clarifier
- Sound to text translator

AI RESPONSE SUMMARY

AI grouped recurring themes from the collected data and highlighted patterns related to dosage estimation, steeping uncertainty, and ritual behavior.

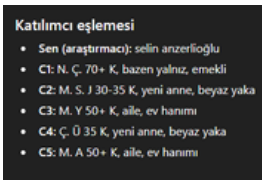
DESIGN DECISION

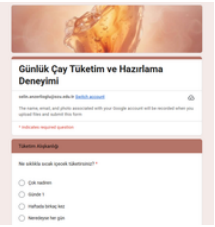
Focus on recurring problems that appeared consistently across multiple participants. Creating Segments and Persona from the outcomes from AI summary about the research. Noted the research and observation notes on Notion page for documenting

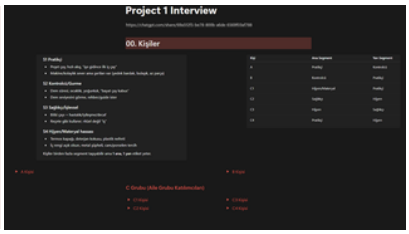
Accepted: The result of the interview, the ratios from the survey, categories. Steeping time uncertainty as a key problem.

Rejected: The interviewer was also talking about their experiences, but "Turboscribe.ai" turned that sound file into a text file, so ChatGPT assumed that everyone talking in the text was the attendee.

Transformed: I sent my handwritten notes first and asked AI to tell me what it sees, and then I sent the transcription text and asked for a double check if they are matching and what AI read and understood from my notes.
I checked my notes and changed the wrong transcription.
Raw interview transcripts → structured insight categories.







FIELD OBSERVATION& EMPATHIZE

Milestone 03

Observation sessions were conducted in a kitchen environment to understand how users interact with hot beverage equipment. Users were observed while preparing tea and coffee, focusing on actions, confusion points, cleaning habits, and safety behavior. Since a machine similar to the intended design was unavailable, coffee machines were examined as a comparable system, including how water is dispensed, containers are placed, heating occurs, cleaning is done, and how mistakes happen during use. The machine was first used without explanation to observe intuitive interaction, followed by operator explanations to reveal hidden mechanisms. Research methods also included interviews, observation, transcript analysis, and a one-hour WhatsApp family group chat analysis; recordings were transcribed with Turboscribe.ai and analyzed with AI, with participants anonymized and coded as A–B–C (interviews) and C1–C4 (group chat).

REFLECTION

The conversation data included overlapping responses and multiple discussion threads. AI helped cluster the responses and make behavioral patterns visible.

SELECTED PROMPT

“Help me organize the insights from my interviews and observation notes + transcripts. I coded attendees for privacy."

AI ROLE

- Clarifier
- Expander

AI RESPONSE SUMMARY

AI grouped recurring themes from the collected data and highlighted patterns related to dosage estimation, steeping uncertainty, and ritual behavior.

SELECTED PROMPT 2

“How can I improve my questions according to UX methods, and what can be my survey questions in categorized”

DESIGN DECISION

Use recurring patterns to define user insights. Learned how to make tea and clean the machine in common kitchens. Also from the analysis of the interface of a coffee machine, i decided to use the same magnet and induction stove mechanism. But change the interface. and transform it for herbal tea.

Accepted: Use recurring patterns to define user insights.
Differences between careful and practical tea preparation behaviors. The mechanics, visual and sound feedback mechanisms. Google forms genenrated extra survey questions.

Rejected: Minor personal preferences that did not affect the preparation process. Usingthe same cup icon for users cup size selection on the digital screen, it creates confusion. The options form the survey was random, i changed them in the context of the information i want to get.
Transformed: Unstructured conversation transcripts → categorized behavioral insights**.** Themed and leveled survey questions to keep atendees in the test and not exit the survey, as a trick.

AI PSYCHOLOGICAL REFLECTION LOG (DÜŞ)

Milestone 04

During this project, I introduced DÜŞ (Thought Log) to document the psychological side of my design process. After I get stuck mentally in my design project, feeling bad.
Design processes often involve moments of confusion, overload, and creative blockage. I used AI as a reflection and thinking support tool to organize these moments and return to the design task.
Selected reflections are included in my logbook as coded DÜŞ entries, showing how AI supported thinking, reflection, and decision-making, rather than generating the design itself.

SELECTED PROMPT

I will use this space like a diary to share my thoughts and blocks during a delayed design process. Your role is to analyze what I write and help me return to the process with balanced support, sometimes by boosting confidence and sometimes by grounding me in reality.

AI ROLE

- Reframer

REFLECTION

AI was used as a thinking partner to help organize thoughts and overcome creative blocks.

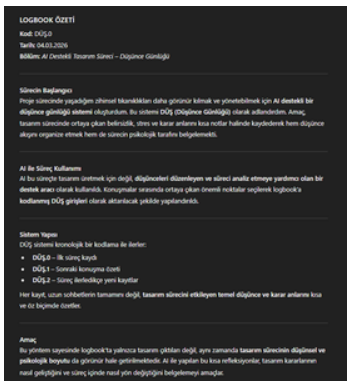
AI RESPONSE SUMMARY

Sürecin Başlangıcı
Proje sürecinde yaşadığım zihinsel tıkanıklıkları daha görünür kılmak ve yönetebilmek için AI destekli bir düşünce günlüğü sistemi oluşturdum. Bu sistemi DÜŞ (Düşünce Günlüğü) olarak adlandırdım. Amaç, tasarım sürecinde ortaya çıkan belirsizlik, stres ve karar anlarını kısa notlar halinde kaydederek hem düşünce akışını organize etmek hem de sürecin psikolojik tarafını belgelemekti.

...

DESIGN DECISION

AI was used as a thinking and reflection tool rather than a design generator. This logbook can be used for the final logbook since it includes all the design process daily, including inner thoughts and feelings, motivations.



LOGBOOK ÖZETİ
Kod: DÜŞ.0
Tarih: 04.01.2026
Bölüm: *AI Destekli Tasarım Süreci – Düşünce Günlüğü*

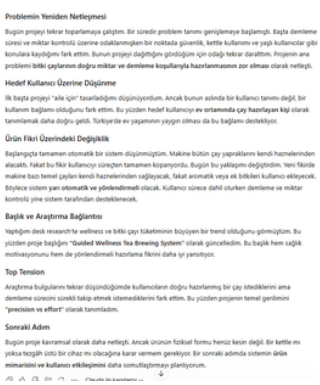
Sürecin Başlangıcı
Proje sürecinde yaşadığım zihinsel tıkanıklıkları daha görünür kılmak ve yönetebilmek için **AI destekli bir düşünce günlüğü sistemi** oluşturdum. Bu sistemi **DÜŞ (Düşünce Günlüğü)** olarak adlandırdım. Amaç, tasarım sürecinde ortaya çıkan belirsizlik, stres ve karar anlarını kısa notlar halinde kaydederek hem düşünce akışını organize etmek hem de sürecin psikolojik tarafını belgelemekti.

AI ile Süreç Kullanımı
AI bu süreçte tasarım üretmek için değil, **düşünceleri düzenleyen ve süreci analiz etmeye yardımcı olan bir destek aracı** olarak kullanıldı. Konuşmalar sırasında ortaya çıkan önemli noktalar seçilerek logbook'a **kodlanmış DÜŞ girişleri** olarak aktarılacak şekilde yapılandırıldı.

Sistem Yapısı
DÜŞ sistemi kronolojik bir kodlama ile ilerler:
- **DÜŞ.0** – İlk süreç kaydı
- **DÜŞ.1** – Sonraki konuşma özeti
- **DÜŞ.2** – Süreç derlendikçe yeni kayıtlar

Her kayıt, uzun sohbetlerin tamamını değil, **tasarım sürecini etkileyen temel düşünce ve karar anlarını** kısa ve öz biçimde özetler.

Amaç
Bu yöntem sayesinde logbook'ta yalnızca tasarım çıktıları değil, aynı zamanda **tasarım sürecinin düşünsel ve psikolojik boyutu** da görünür hale getirilmektedir. AI ile yapılan bu kısa refleksiyonlar, tasarım kararlarının nasıl geliştiğini ve süreç içinde nasıl yön değiştiğini belgelemeyi amaçlar.



Problemin Yeniden Netleşmesi
Bugün projeyi tekrar toparlamaya çalıştım. Bir süredir problem tanımı genişlemeye başlamıştı. Başta demleme süreci ve miktar kontrolü üzerine odaklanmışken bir noktada güvenlik, kettle kullanımı ve yaşlı kullanıcılar gibi konulara kaydığımı fark ettim. Bunun projeyi dağıttığını gördüğüm için odağı tekrar daraltttım. Projenin ana problemi **bitki çaylarının doğru miktar ve demleme koşullarıyla hazırlanmasının zor olması** olarak netleşti.

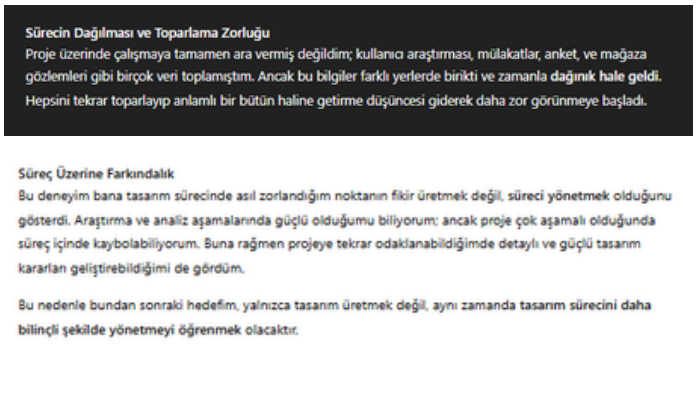
Hedef Kullanıcı Üzerine Düşünme
İlk başta projeyi "aile için" tasarladığımı düşünüyordum. Ancak bunun aslında bir kullanıcı tanımı değil, bir kullanım bağlamı olduğunu fark ettim. Bu yüzden hedef kullanıcıyı **ev ortamında çay hazırlayan kişi** olarak tanımlamak daha doğru geldi. Türkiye'de ev yaşamının yaygın olması da bu bağlamı destekliyor.

Ürün Fikri Üzerindeki Değişiklik
Başlangıçta tamamen otomatik bir sistem düşünmüştüm. Makine bütün çay yapraklarını kendi haznelerinden alacaktı. Fakat bu fikir kullanıcıyı süreçten tamamen kopanyordu. Bugün bu yaklaşımı değiştirdim. Yeni fikirde makine bazı temel çayları kendi haznelerinden sağlayacak, fakat aromatik veya ek bitkileri kullanıcı ekleyecek. Böylece sistem **yarı otomatik ve yönlendirmeli** olacak. Kullanıcı sürece dahil olurken demleme ve miktar kontrolü yine sistem tarafından desteklenecek.

Başlık ve Araştırma Bağlantısı
Yaptığım desk research'te wellness ve bitki çayı tüketiminin büyüyen bir trend olduğunu görmüştüm. Bu yüzden proje başlığını **"Guided Wellness Tea Brewing System"** olarak güncelledim. Bu başlık hem sağlık motivasyonunu hem de yönlendirmeli hazırlama fikrini daha iyi yansıtıyor.

Top Tension
Araştırma bulgularını tekrar düşündüğümde kullanıcıların doğru hazırlanmış bir çay istediklerini ama demleme sürecini sürekli takip etmek istemediklerini fark ettim. Bu yüzden projenin temel gerilimini **"precision vs effort"** olarak tanımladım.

Sonraki Adım
Bugün proje kavramsal olarak daha netleşti. Ancak ürünün fiziksel formu henüz kesin değil. Bir kettle mı yoksa tezgâh üstü bir cihaz mı olacağına karar vermem gerekiyor. Bir sonraki adımda sistemin **ürün mimarisini ve kullanıcı etkileşimini** daha somutlaştırmayı planlıyorum.



Sürecin Dağılması ve Toparlama Zorluğu
Proje üzerinde çalışmaya tamamen ara vermiş değildim; kullanıcı araştırması, mülakatlar, anket, ve mağaza gözlemleri gibi birçok veri toplamıştım. Ancak bu bilgiler farklı yerlerde birikti ve zamanla **dağınık hale geldi.** Hepsini tekrar toparlayıp anlamlı bir bütün haline getirme düşüncesi giderek daha zor görünmeye başladı.

Süreç Üzerine Farkındalık
Bu deneyim bana tasarım sürecinde asıl zorlandığım noktanın fikir üretmek değil, süreci yönetmek olduğunu gösterdi. Araştırma ve analiz aşamalarında güçlü olduğumu biliyorum; ancak proje çok aşamalı olduğunda süreç içinde kaybolabiliyorum. Buna rağmen projeye tekrar odaklanabildiğimde detaylı ve güçlü tasarım kararları geliştirebildiğimi de gördüm.

Bu nedenle bundan sonraki hedefim, yalnızca tasarım üretmek değil, aynı zamanda tasarım sürecini daha bilinçli şekilde yönetmeyi öğrenmek olacaktır.

Accepted: AI-assisted reflection and support, logbook styled written DÜŞ log. Logical assistance, not a romantic one.
Rejected: AI-generated design ideas. And be Praised.
Transformed: Emotional process → documented design reflection.

USER JOURNEY AND SCENARIO

Milestone 05

The tea preparation process was mapped into stages to identify where users experience difficulties.

23-year-old tea drinker living in a family home, drinks tea daily.

	ACCESS	CHOOSE	BREW	DRINK / EXPER.	CLEAN
Touchpoints	• Kitchen • Kettle or teapot • Tea box • Cup	• Tea container • Herbal tea mixes • Spoon / tea bag	• Kettle • Brewing vessel • Tea leaves	• Tea cup • Table / environment	• Tea residue • Teapot • Sink
Actions	1. Realizes they want tea 2. Fills the kettle with water 3. Turns on the kettle	1. Chooses which tea to drink 2. Adds tea into cup or teapot	1. Adds hot water 2. Waits for brewing	1. Checks tea color 2. Drinks tea	1. Disposes tea leaves 2. Cleans equipment
Painpoints	• Waiting for water to boil • Water taste issues (limescale / metallic taste)	• Not knowing the correct dosage • Uncertainty when mixing herbs	• Brewing time is unclear • Users keep checking the tea	• Inconsistent taste • Tea cools down quickly	• Cleaning effort • Limescale buildup
	(Survey: 59% say water quality strongly affects tea taste)	*(Survey: 43% estimate tea dosage visually)* *(Approx. 1.5 g base tea per cup)*	*(Survey: 54% estimate brewing time visually)*	*Survey: 81% say tea color is important*	*Survey: 78% value easy cleaning*
Emotions	🙂 *Anticipation* 🙂 *Waiting*	🙁 *Uncertainty* 🙂 *Curiosity*	😐 *Waiting* 🙁 *Doubt about readiness*	🙂 *Relaxation* 🙂 *Enjoyment*	😒 *Annoyance*
Opportunities	• Temperature control • Water filtration • Faster heating process	• Dosage guidance • Measuring system • Herbal combination guidance	• Brewing timer • Visual readiness indicator • Assisted brewing process	• Heat retention • Visual brewing indicator	• Removable filter • Easy-clean design

KEYWORDS

- Tea ritual
- Daily routine
- Health perception
- Functional beverage

SELECTED PROMPT

"Help me structure the tea preparation process into stages."

REFLECTION

This made it easier to identify where design intervention is needed.

AI RESPONSE SUMMARY

AI organized the preparation process into structured interaction stages.

AI ROLE

- Clarifier
- Critique

DESIGN DECISION

Focus design improvements on the brewing stage.

Accepted: Brewing stage as intervention point.
Rejected: Redesigning the entire kitchen environment.
Transformed: User actions → interaction structure.

The tea preparation process was mapped into stages to identify where users experience difficulties.

INTERACTION FLOW

User selects a tea blend recipe
The system dispenses the required base tea
The screen displays additional ingredients to add
The user places herbs into a visible brewing chamber
The system controls steeping time and brewing
The finished tea is dispensed

PRODUCT CONCEPT

Milestone 06

WORKING SCENARIO

The user selects two tea recipes from the interface.
The machine dispenses the base tea, while the user adds extra herbs manually.
Water is heated internally and delivered into two transparent brewing chambers.
The system controls steeping time and temperature, then filters and dispenses the tea into two cups.

APPROXİMATE CAPACİTY

1. Water reservoir: 1.5-2 L
2. Each brewing chamber: 250-300 ml
3. Cup output: 1 cup per chamber / 2 cups in total
4. Dry tea storage: enough for several uses of base tea

PRODUCT STRUCTURE

- 1 water reservoir
- 1 heating unit
- 2 base tea storage compartments
- 2 manual herb inputs
- 2 transparent brewing chambers
- 2 filtered outputs
- 1 digital interface

SELECTED PROMPT

"I designed a dual herbal tea brewing machine where users add herbs manually while the system controls water temperature and steeping time.
Can you help me structure a realistic working system explaining how water heating, brewing chambers, filters, and dispensing would work step by step?"

HOW THE PRODUCT WORKS

The system combines machine precision with user participation.

Step 1 — Tea selection

The user selects two tea recipes from the digital interface.
Each side can prepare a different tea at the same time.
Example:
- Left chamber: green tea + chamomile
- Right chamber: black tea + mint

Step 2 — Base tea dispensing

The machine releases the required amount of base tea from its internal storage compartments into each brewing chamber.
These base teas may include:
- black tea
- green tea
- white tea

Step 3 — Manual herb addition

The user adds extra herbal ingredients manually through the top openings.
This step keeps the user involved in the ritual and allows personalization.

Step 4 — Water heating

Clean water is drawn from the side reservoir, sent through the internal heating unit, and heated to the required temperature for the selected tea.

Step 5 — Controlled brewing

The heated water is directed into the transparent brewing chambers.
Tea leaves and herbs steep inside the glass chambers for the correct duration.
The user can visually follow:
- leaf movement
- tea color
- brewing progress

Step 6 — Steeping control

The system controls steeping time automatically.
This prevents:
- over-steeping
- bitterness
- inconsistent taste

Step 7 — Filtering and serving

When brewing is complete, the tea passes through the internal filter and is dispensed into the cups placed below.
Each chamber serves its own cup separately.

KEYWORDS

- Guided brewing
- Semi-automated preparation
- Ritual preservation
- Steeping control
- Dual tea brewing
- Visible brewing process
- Precision + user participation

AI ROLE

- Clarifier: helped structure the internal workflow of the machine.
- Expander: suggested realistic internal components and brewing sequence.

AI RESPONSE SUMMARY

- AI helped organize the product into a clear functional system componenets.
- It also structured the brewing sequence from ingredient input to final serving, making the working principle easier to communicate in the design presentation.

REFLECTION

At this stage I realized that my sketches focused mostly on the form and ritual experience, but the internal working mechanism was not yet clear.

DESIGN DECISION

- Users manually add herbal ingredients
- The machine controls water temperature and steeping time
- Brewing occurs in transparent chambers so the process remains visible
- Two teas can be brewed simultaneously
- The machine assists precision without removing the ritual

ACCEPTED

- Semi-automated brewing concept
- Dual brewing chambers
- Visible glass brewing system
- Automated steeping control

REJECTED

- Fully automated capsule-style tea machine
- Hidden brewing process
- Complex internal ingredient mixing systems

These options removed the ritual and reduced user engagement.

TRANSFORMED

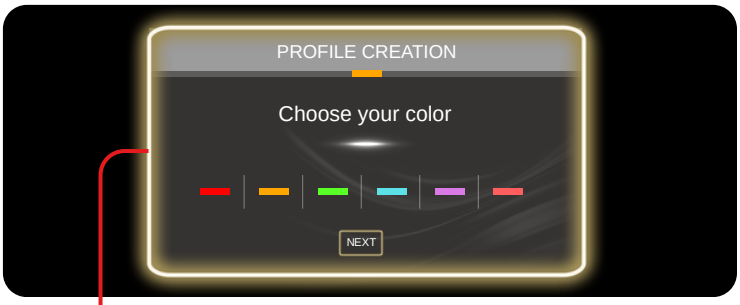
Initial idea:
"Automatic tea machine"
↓
Refined concept:
Guided brewing system that assists precision while keeping the ritual of tea preparation visible and interactive.

INTERFACE

Milestone 07

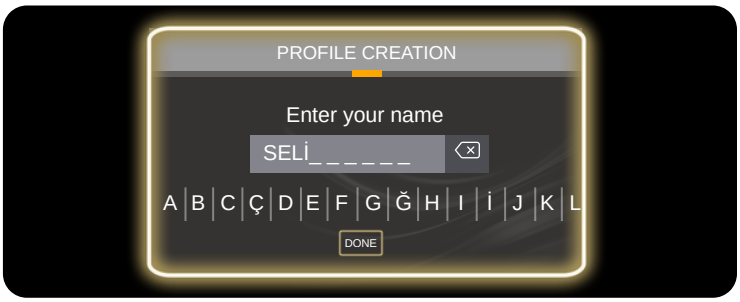
The interface guides users through the tea preparation process in a simple, step-by-step flow. Users first create a personal profile by selecting a color and entering their name, allowing the machine to visually distinguish different users through color-coded indicators. After creating a profile, users choose their tea combination by selecting base teas and optional herbal ingredients from the screen. The interface then displays common tea options and brewing selections while keeping the interaction clear and minimal, helping users prepare tea without confusion while still remaining involved in the ritual of tea making.

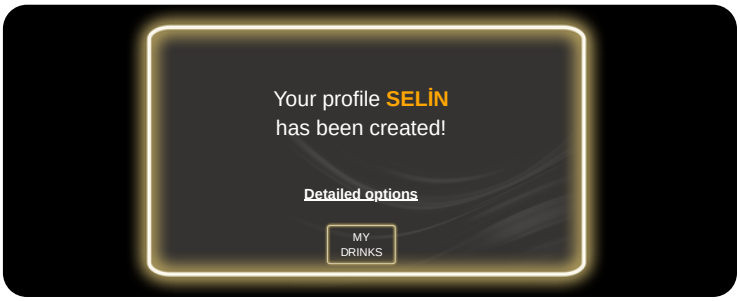
STEP 1



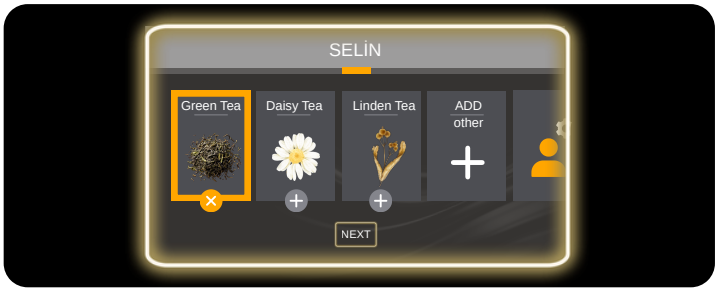
The color of the profile chosen as a led light

STEP 2





Profile Color options



Most common teas
Up to 30 profiles

SKETCHING & FORM EXPLORATION

Milestone 08